

The SPO portfolio tree: model v1

The model and its training, with the equations; then what the planted-threshold experiment says

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Outline

What we do, and how we test it

The model: the tree, the decision problem, the loss, the training

Model v1: the split-search options and the frozen configuration

The planted-threshold experiment: recovery, performance, trading rate, what the options bought, v1 on 100 datasets, resolution

A fair assessment, and what comes next

What we do, and how we test it

Problem

- weights of n assets every h days, from features x_t , with fees
- a decision problem, not a forecast

Model

- small tree $\rightarrow$ leaf scores $\rightarrow$ mean–variance optimizer with fees
- loss: decision regret through the optimizer
- splits confirmed on the tree's own traded path
- Out of sample Sharpe pruning

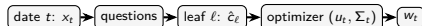
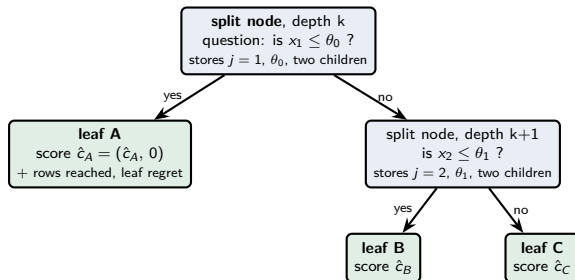
Test

- synthetic market, one feature x , planted threshold d
- benchmark: *ideal rule* = optimizer fed $\mathbb{E}[r \mid x]$
- constant volatility; asset vs cash; weekly unless stated

Model v1

- frozen 24 September 2026
- $\approx 15,000$ fitted trees behind the numbers
- validated on 100 datasets

The tree: what a node stores



- tree = nested questions "is $x_j \leq \theta$?", ending in leaves
- **split node**: feature j , threshold θ , two children; no score
- **leaf**: one score vector $\hat{c}_\ell = (\hat{c}_\ell, 0)$; rows reached; leaf regret
- holdings u_t , weights w_t , returns: per date, not per node
- depth $\leq 3 \Rightarrow \leq 8$ leaves $\Rightarrow \leq 8$ scores
- growth: leaf $\rightarrow$ split node + 2 leaves; pruning: the reverse
- use: $x_t \rightarrow$ leaf $\rightarrow \hat{c}_\ell \rightarrow$ optimizer(u_t, Σ_t) $\rightarrow w_t$

The decision: mean–variance with proportional fees

Notation

- u : holdings before the decision,

$$u_{t+1} = \frac{w_t \circ (1 + r_t)}{1 + r_t^\top w_t}, \quad u_1 = \mathbf{1}/n$$
- c : the tree's scores; cash = 0 (cash earns nothing here + only the score differences matter)
- Σ : h -day covariance from the last 180 days;
 $\sigma_h^2 = h\hat{\sigma}^2$
- f : 3 bp for the 11asset / 0 cash; $\lambda = 1$
- $\mathcal{W} = \{w : 0 \leq w_i \leq 1, \sum_i w_i = 1\}$

$$w^*(c; u, \Sigma) = \arg \max_{w \in \mathcal{W}} \underbrace{c^\top w}_{\text{gain}} - \underbrace{f^\top |w - u|}_{\text{fees}} - \underbrace{\lambda w^\top \Sigma w}_{\text{risk}}$$

Asset vs cash (w = asset weight, $\kappa = 2\lambda\sigma_h^2$):

$$w^* = \begin{cases} \min\{(c - f)/\kappa, 1\} & c - \kappa u > f \text{ (buy)} \\ \max\{(c + f)/\kappa, 0\} & c - \kappa u < -f \text{ (sell)} \\ u & |c - \kappa u| \leq f \text{ (no trade)} \end{cases}$$

v1: $\kappa = 0.1\%$ (weekly, $\sigma = 1\%$ /day); $w : 0 \rightarrow 1$ for $c : 0.03\% \rightarrow 0.13\%$ /week; leaf scores $\pm 0.3\text{--}0.5\% \Rightarrow w \in \{0, 1\}$; interior positions (continuous sizing) need $\lambda \geq 5$

Properties

- strictly concave $\Rightarrow$ unique w^*
- $w^*(c + \alpha \mathbf{1}) = w^*(c)$: only score differences matter
- exact solution: 5 states per weight (0, cap = $\bar{w}$, u_i , buy, sell) $\Rightarrow 5^n$ linear systems, best feasible one

The loss: decision regret; the score of a leaf

$$U_t(w) = r_t^\top w - f^\top |w - u_t| - \lambda w^\top \Sigma_t w$$

$$\text{Reg}_t(c) = U_t(w^*(r_t; u_t, \Sigma_t)) - U_t(w^*(c; u_t, \Sigma_t)) \geq 0$$

hindsight decision – decision on c (given the same holdings u_t)

$$\hat{c}_\ell = \arg \min_{c \in [-b, b]^n} \sum_{t \in \ell} \text{Reg}_t(c), \quad b = 10\%$$

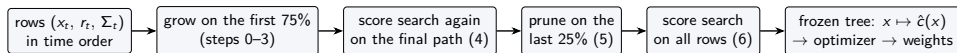
Leaf score

- one regret per date, summed over the leaf's dates
- bounded coordinate search: 3 passes $\times$ 7 values
- not a forecast: same regret for any score on the right side

Consequences

- fees $\Rightarrow$ regret depends on $u_t \Rightarrow$ holdings from a replay of the tree
- only differences: relative returns, asset vs cash $\Rightarrow$ help to choose the right features
- same Σ_t as the optimizer: unseen risk not learnt

Training: replay, two growth gates, out of sample pruning



0. root score from $u_t = 1/n$; **replay** (trade the scores date by date from u_1) $\Rightarrow u_t$, hindsight utilities, Sharpe S ; holdings frozen for the round
1. **regret gate**, per leaf: score search again from current u_t (yardstick L_ℓ); candidates; children of the 3 best; keep iff $\Delta = L_\ell - L_{\text{left}} - L_{\text{right}} > 10^{-6}$ per row
2. **Sharpe gate**: rank all kept splits by Δ ; whole-tree replay of the first; accept iff $S' > S$, else next; one per round; none $\Rightarrow$ stop
3. replay the new tree $\Rightarrow$ new u_t ; back to 1 (depth ≤ 3 ; ≥ 20 rows and $\geq 10\%$ of the parent per leaf)
4. score search again on every leaf, final path; kept iff S does not fall
5. **pruning**, bottom-up, on the last 25% (never used for growth): with vs without the split, both replayed; $g_t = \text{Reg}_t(\text{without}) - \text{Reg}_t(\text{with})$; keep iff $\bar{g} > 10^{-6}$, $\bar{g} > k_p \text{se}(\bar{g})$ [$k_p = 2$ below the root, 0 at the root], checking Sharpe $\uparrow$
6. score search again on all training rows

Gates 1–2: in-sample. Gate 5: out-of-sample.

Split-search options: quantile candidates, refinement, soft splits

Common: admissible candidates $\theta_{(1)} < \dots < \theta_{(K)}$ (each side $\geq m_* = \max(m, \lceil \varphi N \rceil)$ rows; $m = 20$, $\varphi = 10\%$); cheap screen

$$Q(\theta) = \sum_{t \in I} \text{Reg}_t(c_{\text{side}(t)}(\theta)), \quad c_{\text{left}}(\theta) = \text{clip}(\text{mean}\{r_t : x_t \leq \theta\}, \pm b), \text{ idem right}$$

Deciles by rank + refinement

- keep ranks $\text{round}(q(K-1))$, $q = 0.1, \dots, 0.9$: 9 screens instead of $K \approx N - 2m_*$
- refinement: screen every candidate between the two deciles around the best one, θ_o ; best of them $\theta_\bullet$
- replace iff $Q(\theta_o) - Q(\theta_\bullet) > k \text{ se}$, $\text{se} = \sqrt{|I|} \text{std}_t(\delta_t)$, $k = 1$
- 3 best survivors $\rightarrow$ full fit of the children

Soft split (off in v1: $\kappa = 0$)

$$z_k = \frac{Q(\theta_k) - \min_I Q(\theta_I)}{\text{se}_k}, \quad \pi_k = \frac{e^{-z_k/\kappa}}{\sum_I e^{-z_I/\kappa}}$$

- all candidates kept, weighted; Gaussian variant $e^{-z^2/2\kappa^2}$
- membership: left with $\mu(x) = \sum_{k: \theta_k \geq x} \pi_k$, right with $1 - \mu(x)$; weights $\omega_{t\ell}$ multiply down the tree
- leaf fit: $\min_c \sum_t \omega_{t\ell} \text{Reg}_t(c)$; row score $\hat{c}(x_t) = \sum_\ell \omega_{t\ell} \hat{c}_\ell$: a ramp, not a jump
- $\kappa \rightarrow 0$: plain split

Model v1: the frozen configuration (24 September 2026)

Choice	v1	Why (section of the experiments report)
risk aversion λ	1	positions 0/1: regime classifier, not sizer (8)
fee	3 bp asset, 0 cash	market-sized; 10 bp starves fast rates (8)
decision rate	weekly ($h = 5$); 2-day as alternative	daily $\approx$ weekly; monthly loses (4)
depth	≤ 3	depth alone changes nothing (8)
leaf minimum	20 decisions and 10% of the node	tail leaves at 0.7% of the data (7)
candidate thresholds	deciles by rank + refinement (1 s.e.)	= base Sharpe, steadier thresholds (9, 12)
soft splits	off	never raise the Sharpe at $\lambda = 1$ (9, 12)
growth gates	regret reduction, then replayed Sharpe $\uparrow$	kept as designed; v2 revisits
pruning	last 25%; hurdle 2 s.e. below the root, none at the root	out-of-sample judge (7); removes spurious leaves at no cost (12)
covariance	180-day window, $\Sigma_t = h\hat{\sigma}_t^2$	20-day window: noisier than timely (regime report)
conventions	cash column; all-cash Sharpe = 0; $ \hat{c} \leq 10\%$	(7, model document)

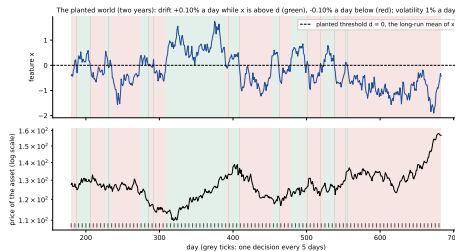
Base of every experiment = same pipeline, every option off: depth 2, all candidates screened, plain pruning.

The test-bed: a market with a planted answer (constant volatility)

$$x_{t+1} = m + e^{-k}(x_t - m) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, s^2(1 - e^{-2k}))$$

$$\log \frac{S_{t+1}}{S_t} = \mu_t - \frac{1}{2}\sigma^2 + \sigma\epsilon_t, \quad \mu_t = \pm e\sigma \text{ as } x_t \gtrless d$$

σ	1% a day (16%/yr), both regimes
e (edge)	0.10, 0.05, 0.02 = $\pm 25, \pm 13, \pm 5\%$ /yr
k	0.02/day (half-life 35 d); 0.01 to 0.10
m, s^2, d	0, 1, 0: threshold at the mean of x
h	1, 2, 5, 10, 21 days
data	15 years training, 5 years test, daily

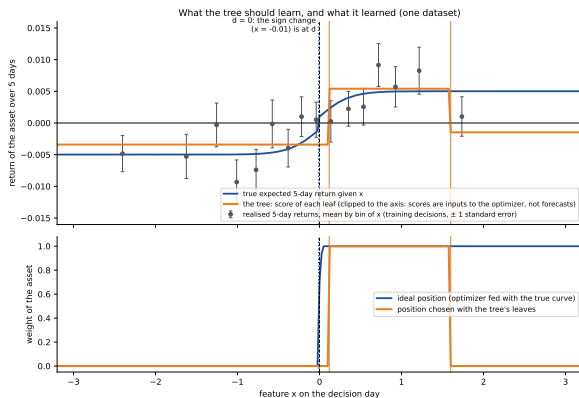


Measured (40–100 datasets per cell)

- first threshold vs planted $d = 0$ (x has sd 1)
- annualized test Sharpe: tree vs ideal rule ($\mathbb{E}[r | x]$ fed to the optimizer), rule at d , always the asset
- share of the ideal gain captured

Second market (volatility follows a threshold too): own report; one line at the end.

One fitted tree (dataset 42, weekly, edge 0.10)



Drawn (x on the decision day)

- blue: true $\mathbb{E}[r \mid x]$, 5-day; zero at $x = -0.01$
- grey: realized 5-day returns, mean by bin of x , ± 1 s.e.
- orange: the tree's score per leaf: -0.35% ($x \leq 0.12$), $+0.55\%$ (to 1.60), -0.15% (above)
- bottom: asset weight from a 50/50 portfolio, fed the curve (ideal) or the score (tree)

Key points

- root threshold 0.12 vs planted 0
- scores $\neq$ curve, and need not be: both outside the 0.1% window $\Rightarrow$ same positions as the ideal rule up to 1.60
- leaf above 1.60: spurious (base pruning); cash where the ideal rule stays invested; removed by the v1 hurdle
- ideal rule = a step too ($\lambda = 1$)

Result 1: the threshold is recovered at market sizes (E2, base pipeline)

edge	h	split	threshold	annual. test Sharpe		
		on x (%)	within 0.25	tree	ideal	gain
0.10	1	97	95%	0.80	1.03	76%
	5	94	81%	0.73	0.88	79%
	21	81	60%	0.54	0.68	75%
0.05	1	82	60%	0.12	0.49	25%
	5	78	53%	0.24	0.43	51%
	21	62	34%	0.17	0.35	43%
0.02	5	54	26%	-0.03	0.15	-3%

- row = 100 datasets (50 at edge 0.02), threshold planted at 0
- split on x: trees keeping a split (else: no split at all)
- within 0.25: first threshold within 0.25 sd of x from 0
- Sharpe: mean over datasets, s.e. ≈ 0.04
- gain: tree's excess test return over the no-split tree, as a share of the ideal rule's

- $\pm 25\%/yr$: **found at every rate**; 95% within 0.25 daily, 60% monthly; 75–79% of the ideal gain
- $\pm 13\%/yr$: **graceful degradation**; 62–82% split; 25–51% of a small gain (ideal 0.35–0.49)
- $\pm 5\%/yr$: **nothing to find in 15 years**; half no split, half at random; ideal rule ≈ 0.15 ; finding nothing = right answer
- daily $\approx$ weekly; monthly loses (regime changes inside the holding period)

Result 2: the strategy's performance out of sample (E2, base pipeline)

edge	h	annualized test Sharpe ratio				gain captured
		tree	ideal rule	rule at d	the asset	
0.10	1	0.80	1.03	1.04	-0.02	76%
	2	0.86	0.97	0.98	-0.02	85%
	5	0.73	0.88	0.88	-0.02	79%
	10	0.63	0.78	0.78	-0.02	78%
	21	0.54	0.68	0.66	-0.02	75%
0.05	1	0.12	0.49	0.49	-0.02	25%
	2	0.21	0.46	0.46	-0.02	44%
	5	0.24	0.43	0.42	-0.02	51%
	10	0.17	0.39	0.38	-0.02	43%
	21	0.17	0.35	0.34	-0.02	43%

- 100 datasets per row; s.e. ≈ 0.04
- ideal rule: optimizer fed $\mathbb{E}[r \mid x]$; rule at d : asset iff $x \geq d$; the asset: always invested
- turnover per weekly decision (edge 0.10): tree 0.25, ideal 0.28 $\Rightarrow$ fees $\approx 0.4\%/yr$

- $\pm 25\%/yr$: **75–85% of the attainable gain** at every rate (0.73 vs 0.88 weekly; 0.86 vs 0.97 at 2 days)
- **rule at $d = \text{ideal rule}$** : at $\lambda = 1$ the threshold is all there is to learn
- lost: threshold error (0.07–0.25) + spurious leaves ≈ 0.15 Sharpe weekly
- $\pm 13\%/yr$: **25–51% of a small gain**: at the limit of what 15 years identify
- always the asset: ≈ 0 ; the value is in the timing

Trading rate against the speed of the feature (edge 0.10, 50 datasets per cell)

half-life of x	daily		weekly		monthly	
	tree	ideal	tree	ideal	tree	ideal
69 days ($k = 0.01$)	0.63	1.09	0.84	0.97	0.68	0.84
35 days ($k = 0.02$)	0.80	1.03	0.73	0.88	0.54	0.68
14 days ($k = 0.05$)	0.69	0.98	0.65	0.75	0.30	0.49
7 days ($k = 0.10$)	0.77	0.96	0.45	0.60	0.18	0.34

annualized test Sharpe ratios, tree and ideal rule

- daily $\approx$ weekly: more decisions = more fees, not more information
- monthly loses, more so for a fast feature: half-life 7 d $\Rightarrow$ 0.18 vs 0.34
- slow feature + daily decisions: over-trading (0.63 vs 1.09)
- $\Rightarrow$ weekly in v1

What the design options bought (E3–E5, E8; 40 datasets per cell, edge 0.10)

setting	test Sharpe of the tree	
	weekly	2-day
base	0.80	0.87
depth 3	0.80	0.89
$\lambda = 5$ / $\lambda = 10$	0.70 / 0.65	0.81 / 0.71
fee 10 bp	0.60	0.53
deciles + refinement	0.82	0.82
soft splits $\kappa = 1$	0.53	0.35
soft splits $\kappa = 0.25$	0.73	0.61
hurdle 2 s.e., every node	0.43	0.47
hurdle 2 s.e., below the root only	0.77	0.74
deciles + refin. + hurdle below root	0.80	0.79

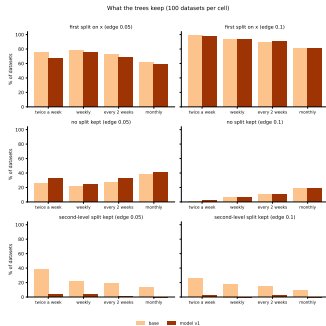
base: depth 2, all candidates, plain pruning; ideal rule 0.91
weekly, 1.00 at 2 days; s.e. ≈ 0.04 ; last row = v1 at depth 2

- **no option beats the base beyond noise**; what is left is threshold noise
- depth 3: nothing (a maximum); $\lambda = 5$ –10: gradual positions, learnt, -0.1 to -0.15 ; 10 bp: fast rates stop trading
- deciles + refinement: = base, 9 candidates instead of hundreds, steadier thresholds $\Rightarrow$ **in v1**
- soft splits: $\kappa = 1$ score in the no-trade band; $\kappa = 0.25$ = base from weekly on, loses at 2 days $\Rightarrow$ **off** (smoothing the score $\neq$ sizing the position at $\lambda = 1$)
- hurdle at every node: loses true root splits (25% block has no power at 2 s.e.) $\Rightarrow$ **off**
- hurdle below the root: second-level leaves 25/10/5/5% $\rightarrow$ 5/0/0/0%, no cost from weekly on $\Rightarrow$ **in v1**

Model v1 on 100 datasets (E9, 800 fits), paired with the base of Result 1

edge	h	leaves		2nd level	within 0.25	Sharpe		v1 — base (2 s.e.)
		base	v1	base → v1	base → v1	base	v1	
0.10	2	2.24	2.00	25 → 2%	92 → 89%	0.86	0.78	−0.08 (0.07)
	5	2.14	1.94	18 → 0%	81 → 82%	0.73	0.73	+0.00 (0.04)
	10	2.03	1.92	14 → 2%	70 → 72%	0.63	0.64	+0.01 (0.05)
	21	1.90	1.81	9 → 0%	60 → 64%	0.54	0.53	−0.01 (0.03)
0.05	2	2.18	1.72	38 → 4%	64 → 57%	0.21	0.18	−0.03 (0.06)
	5	2.01	1.80	21 → 3%	53 → 51%	0.24	0.22	−0.03 (0.05)
	10	1.93	1.69	19 → 1%	47 → 49%	0.17	0.20	+0.03 (0.04)
	21	1.76	1.59	13 → 0%	34 → 42%	0.17	0.14	−0.03 (0.03)

same 100 datasets (seeds 0–99); 2nd level: datasets keeping a second-level split; last column: mean paired difference of annualized test Sharpe, 2 s.e. in brackets; half of the datasets: identical Sharpe



- weekly and slower: v1 = base within noise; 2nd-level leaves gone; 1.9 leaves vs 2.1
- 2-day: -0.08 ± 0.07 (edge 0.10): real 2nd-level splits removed; softer hurdle not run
- depth budget unused ($\leq 1\%$ reach depth 2)
- v1 = base with a cleaner structure, not a higher Sharpe

What can be found at all: resolution against volatility and feature noise (E6)

sweep (weekly)	value	edge	median $ \theta $	tree / ideal
volatility σ , drift $\pm 25\%/yr$	0.4%	0.25	0.05	1.85 / 2.10
	1.0%	0.10	0.12	0.69 / 0.87
	2.0%	0.05	0.39	0.23 / 0.42
	4.0%	0.025	0.95	-0.05 / 0.18
noise τ on x	0	0.10	0.12	0.69 / 0.87
	0.25	0.10	0.19	0.66 / 0.79
	0.5	0.10	0.25	0.45 / 0.66
	1	0.10	0.55	0.26 / 0.48
	2	0.10	1.15	0.09 / 0.29

60 datasets per row; $|\theta|$: first threshold's distance from the planted 0 (sd of x); τ in sd of x ; tree / ideal: annualized test Sharpe

- threshold error $\approx 0.012/e$ down to $e \approx 0.05$; below: a guess
- noise τ on $x \approx$ edge divided by $1 + \tau^2$
- $\tau \leq 0.25$: free; 0.5: error doubles; 1: ideal Sharpe halves
- edge below a few bp/day: unlearnable from 15 years, by any method
- $\Rightarrow$ pre-test every real feature with this protocol

A fair assessment, and what comes next

Holds

- $\pm 25\%$ regime, monthly feature: recovered at every rate; 75–85% of the attainable gain
- $\pm 13\%$: graceful; $\pm 5\%$: correctly nothing
- $v1$ = base with smaller trees, steadier thresholds (100 paired datasets); cost 0.08 at 2 days
- second market (volatility regime): drift threshold found better when the calm regime pays; a variance-only boundary not learnt (lagged covariance in the regret)

Weak or untested

- no option beats the base beyond noise: structure, not Sharpe
- growth Sharpe gate: in-sample, mixes a risk preference into the structure
- one clean feature; synthetic data only; decoys only at an unrealistic edge; depth budget unused

Next (v2)

- growth gate off / replayed regret instead; same 100 datasets
- regime-aware variance for optimizer and regret
- two features + decoys at market-sized edges
- real data; this protocol as the pre-test of every feature